St. Francis Institute of Technology

Department of Information Technology

Academic Year: 2025-2026

Mini Project –Sensor Lab

PROPOSAL PRESENTATION

EVBHARAT

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Introduction

In recent years, the rapid growth of electric vehicles (EVs) and intelligent transportation systems has transformed modern mobility by promoting eco-friendly and efficient travel. However, with the increasing number of vehicles on roads, challenges such as accident prevention, obstacle detection, battery health management, and real-time communication have become critical concerns. Traditional vehicle monitoring systems often operate as separate modules and fail to provide a unified platform for safety, automation, and continuous tracking. To overcome these limitations, **EVBHARAT** is proposed as an intelligent electric vehicle prototype that integrates Bluetooth-based communication, multi-sensor accident detection, battery monitoring, ESP32-CAM object detection, and automation inspired by the Dark Factory concept. This system aims to enhance road safety, improve energy efficiency, and support smarter vehicle decision-making through real-time monitoring and automated alerts.

Problem Statement

Existing systems lack a reliable communication framework that enables secure and efficient interaction between vehicles, infrastructure, and surrounding entities. Conventional RF-based communication often suffers from spectrum congestion and electromagnetic interference, reducing system reliability. This project proposes a Visible Light Communication (VLC) based Vehicle-to-vehicles (V2V) prototype integrated with dark factory automation, where a robotic bot transfers products within a factory while communicating through visible light signals. The system aims to improve communication reliability, coordination, and operational efficiency for smart transportation and automation systems.

Objectives

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- To design and develop a Vehicle-to-Vehicle (V2V) communication system using Visible Light Communication (VLC) for secure and reliable short-range data exchange.
- To implement a multi-sensor accident detection mechanism using accelerometer and vibration sensors for real-time crash and impact detection.
- To develop a predictive battery monitoring model that estimates battery status and supports efficient energy management for vehicle operation.
- To enable secure vehicle communication through light-based data transmission and object detection techniques to minimize interference and enhance communication reliability and data security.

Literature Review

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Serial No.	Research Paper	Description	Limitations
1	**Visible Light Communication for Connected Vehicles: How to Achieve the Omnidirectional Coverage ?**	It investigates the use of VLC in connected vehicle networks and proposes methods to achieve 360° coverage through multiple photodetectors and optimized receiver placement.	• Line-of-sight dependency • Hardware complexity • Night/day variability
2	**Vehicle-to-Everything (V2X) Communications: Enabling Technologies, Challenges, and Opportunities**	It is a survey article on next-generation 6G-enabled Vehicle-to-Everything (V2X) communications that outlines key technologies (e.g., ML, THz, RF/VLC integration), discusses challenges and opportunities, and highlights how future networks can support ultra-reliable, low-latency vehicular connectivity and intelligent transportation systems.	• Security and privacy risks • Standardization gaps • High deployment cost • Technological immaturity

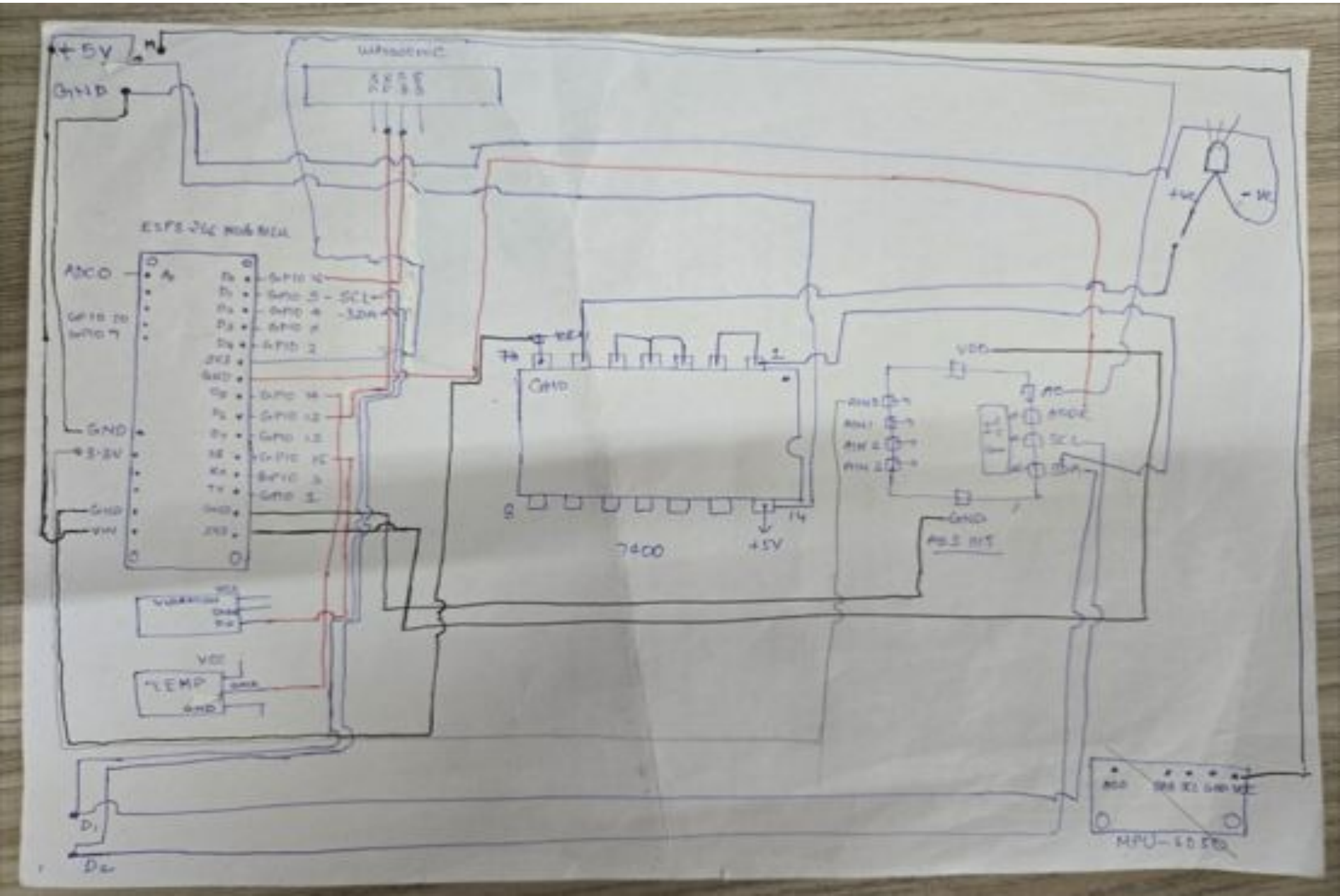
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Literature Review

Serial No.	Research Paper	Description	Limitations
3	**Hybrid RF/VLC Systems: A Comprehensive Survey on Network Topologies, Performance Analyses, Applications, and Future Directions**	It surveys hybrid RF/VLC communication systems, analyzing network architectures, performance, applications, and future research directions, and highlights how combining RF reliability with VLC capacity improves communication robustness.	• Increased system complexity • Higher deployment cost • Interference and noise • Line-of-sight dependence

Block Diagram

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Hardware Requirements

- **Microcontroller & Processing:**ESP32 DevKit V1 is used as the main controller for sensing, processing, and communication. The ESP32-CAM module enables camera-based object detection and monitoring.
- **Communication System:**VLC is implemented using a high-brightness LED as the transmitter and a BPW34 photodiode as the receiver. 7400 NAND gates perform signal modulation, and the ADS1115 ADC converts the photodiode analog signals to digital data.
- **Sensors:**MPU6050 and LX335 accelerometers are used for motion and accident detection. A vibration sensor detects impacts, and a temperature sensor monitors battery temperature.
- **Mobility & Actuation:**An L298 motor driver controls the DC motors responsible for robotic vehicle movement.
- **Power Supply:**The system is powered by three 3.7V Li-ion batteries.
- **Alert Mechanism:**A buzzer provides audio alerts, while LEDs provide visual warnings during system events.

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Software Requirements

Development Environment: Arduino IDE and ESP32 Board Support Package

Programming Language: C / C++ (Embedded programming)

Libraries & Frameworks

- MPU6050 Library (Sensor interfacing)
- I2C & SPI Communication Libraries
- ADC driver libraries
- Camera / Image processing libraries (ESP32-CAM support)

Algorithms & Logic

- Accident detection algorithm (Threshold-based + sensor fusion)
- Battery prediction logic (Software-based estimation model)
- Object detection logic (Computer vision processing)
- VLC modulation & demodulation algorithm
- Intelligent alert generation system
- Emergency decision-making logic

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Conclusion

The EV Bharat project presents a Vehicle-to-Vehicle (V2V) communication system that utilizes Visible Light Communication (VLC) for secure and reliable short-range data exchange. The system integrates multi-sensor accident detection and predictive battery monitoring to enhance vehicle safety and operational efficiency. The concept is further demonstrated through a dark factory automation environment, where a robotic bot transfers products between locations while communicating through visible light signals. Overall, the proposed system demonstrates a practical approach toward developing smarter, safer, and more efficient vehicle communication and automation systems.

References

- *Visible Light Communication*, Researchgate.
- Research papers on **Accident Detection and Alert Systems using GPS** – IEEE Xplore.
- "A Survey on Intelligent Accident Detection and Reporting System for Smart Vehicles," IJERT.
- VLC system design challenges and solutions – IEEE Communications Surveys & Tutorials.
- Datasheets:
  - MPU6050 – InvenSense Product Datasheet
  - ESP32 – Espressif Systems Technical Reference Manual
  - ADS1015 ADC – Texas Instruments Datasheet
  - BPW34 Photodiode – Vishay Semiconductor Datasheet
  - 7400 NAND Gate IC – Texas Instruments / ON Semiconductor Datasheet